

2D Projectile Motion: Launch At Arbitrary Angles

SYNOPSIS

A projectile launched at an angle is really two separate motions running on one clock: a horizontal motion at constant velocity, and a vertical motion in freefall. They don't interact — gravity never touches the horizontal component, and nothing about the horizontal component changes how fast the projectile falls. Splitting $\mathbf{V}_0$ into $V_{0x} = V_0 \cos \theta$ and $V_{0y} = V_0 \sin \theta$ lets us reuse the freefall equations from Note 3 on the vertical component, and constant-velocity equations on the horizontal one. Because only the vertical component ever stops and reverses, the **flight time depends only on V_{0y}** — the horizontal speed is along for the ride. This note works a full numeric example (a launch off a cliff), then sets up the general question N5 finishes: given where you want to land, what angle do you launch at?

PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- PE1 Resolve a launch velocity into components** — $V_{0x} = V_0 \cos \theta$ and $V_{0y} = V_0 \sin \theta$ — and explain why the two are independent.
- PE2 Explain why flight time depends only on V_{0y}** , and derive $t_{\text{flight}} = 2V_{0y}/g$ for a level-ground launch.
- PE3 Derive the max-height formula** $d_{y,\text{max}} = V_{0y}^2/(2g)$ from $v_y^2 = v_{0y}^2 - 2g \cdot d$, and explain why $v_y = 0$ at the apex.
- PE4 Solve a cliff-launch flight-time problem** by writing the vertical position equation and solving the resulting quadratic in t .
- PE5 Work a launch problem end-to-end** — components, flight time, max height, range, and impact velocity (speed and direction).
- PE6 Recognize the general cliff-launch relationship** as a quadratic in $\tan \theta$, and see how it connects to solving for a launch angle in N5.



You already know this

Nothing here is brand new — you're combining three things you've already built:



Resolving a vector (N2 / N2a)

$\cos\theta$ and $\sin\theta$ split any vector into horizontal and vertical pieces. V_0 is no different.



Freefall (N3)

The vertical component obeys the exact same $v = v_0 - gt$ and $d = d_0 + v_0t - \frac{1}{2}gt^2$ you used for straight-up-and-down motion.



Constant velocity (M1)

The horizontal component never changes — no acceleration acts sideways — so $d_x = V_{0x}t$ the whole flight.

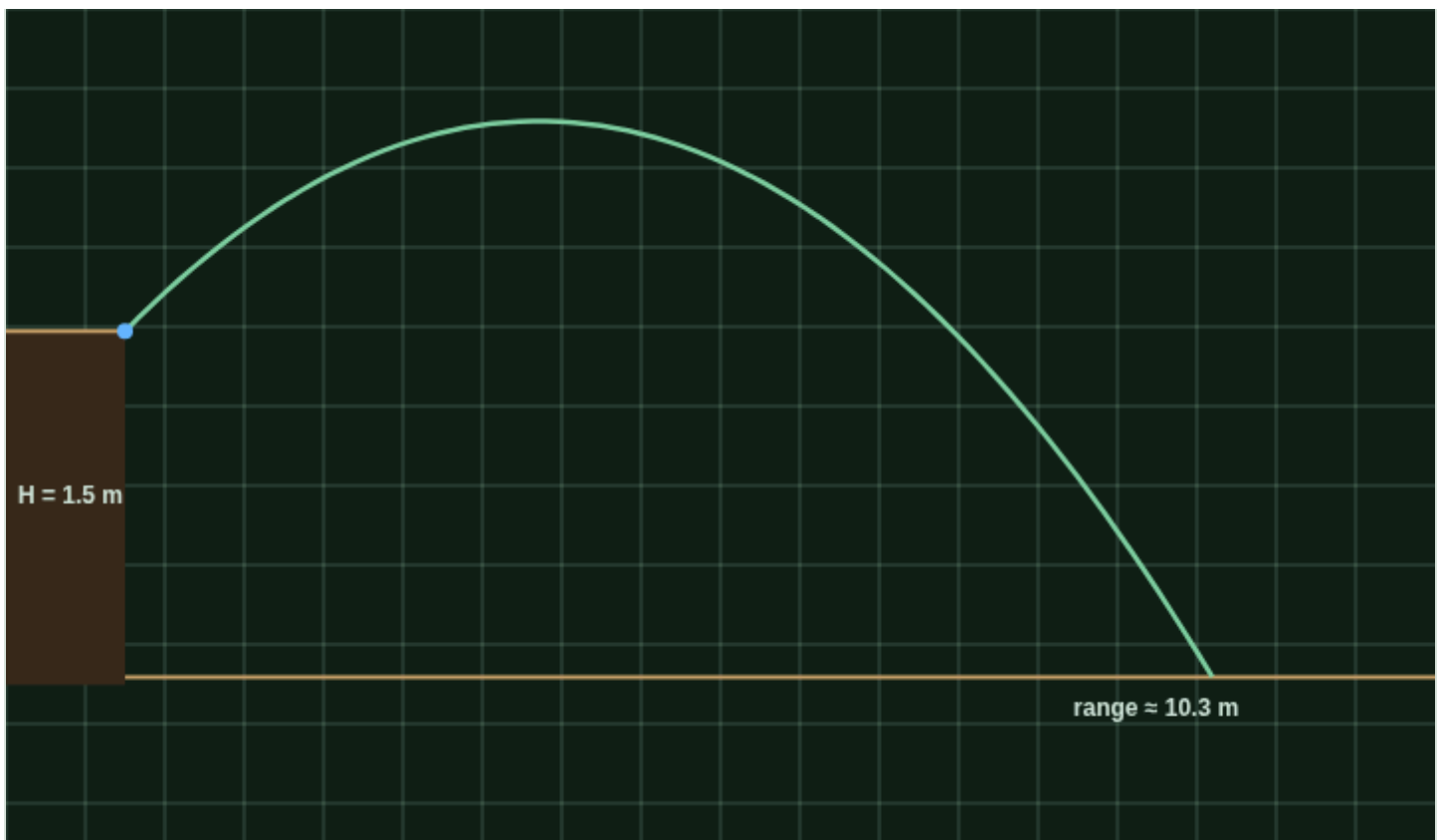
In this note, we work through

- I. How do we split a launch velocity into horizontal and vertical components?
- II. Why does flight time depend only on the vertical component — not the horizontal one?
- III. How far up does the projectile go, and how do we find that height?
- IV. Worked example: given V_0 , θ , and a cliff height, find the flight time, range, and impact velocity.
- V. If we flip the problem — given where we want to land — how do we solve for the launch angle?

Play with it: projectile launcher

Set the launch speed V_0 , launch angle θ , and cliff height H . Press **Launch** to fly the trajectory and watch the readouts update. Notice: raising H only changes when the flight ends, not the shape of the arc on the way up.

► INTERACTIVE · PROJECTILE LAUNCHER



Launch speed V 10.0 m/s Launch angle θ 25° Cliff height H 1.5 m

▶ Launch

↺ Reset

Flight time $\approx 1.13 \text{ s}$

Max height $\approx 0.91 \text{ m}$

Range $\approx 10.3 \text{ m}$

Impact speed $\approx 11.4 \text{ m/s}$

Flight time only changes when V_0 or θ change — try dragging H alone and watch t_{flight} react anyway, since a taller cliff still means more falling to do.

Predict first: two balls are launched off the same cliff with the same speed V_0 — one at 20° above horizontal, the other at 70° above horizontal. Which one stays in the air longer?

The 20° ball

The 70° ball

They land at exactly the same time

Not enough information

Watch: working through the note

WALKTHROUGH VIDEO • ~12 MIN

Video embed goes here — print the guided notes below before you press play.

Setting up the components

A projectile is launched with speed V_0 at angle θ above horizontal. As always, we resolve it into a horizontal and a vertical piece:

$$V_{0x} = V_0 \cos \theta \quad V_{0y} = V_0 \sin \theta$$

From this point on, treat the two components as two separate 1D problems sharing the same clock. Horizontally, nothing accelerates the projectile — V_x stays equal to V_{0x} for the entire flight. Vertically, gravity acts the whole time, exactly like the freefall you already know:

$$V_y = V_{0y} - gt$$

Why flight time depends only on V_{0y}

At the very top of the arc — the **apex** — the projectile is momentarily moving purely horizontally, so $v_y=0$ there. For a launch and landing at the *same height*, the trip down mirrors the trip up: the projectile returns to launch height moving at $-V_{0y}$, the same speed it left with, just flipped. Setting $V_y=-V_{0y}$ in the equation above:

$$-V_{0y} = V_{0y} - gt_{\text{flight}} \Rightarrow 2V_{0y} = gt_{\text{flight}}$$

$$t_{\text{flight}} = 2V_{0y} / g \quad (\text{level launch and landing})$$

Notice: V_{0x} never appears. Two projectiles launched at the same angle but different speeds spend different amounts of time in the air; two projectiles launched at the same V_{0y} but different V_{0x} — meaning different angles and speeds — spend exactly the same time in the air. Horizontal motion just carries the projectile sideways for however long the vertical motion allows.

How high does it go?

Use the no-time kinematic equation on the vertical component, with $a=-g$ and $v_y=0$ at the apex:

$$v_y^2 = v_{0y}^2 - 2g \cdot d_{y,\text{max}} \Rightarrow 0 = (V_0 \sin \theta)^2 - 2g \cdot d_{y,\text{max}}$$

$$d_{y,\text{max}} = (V_0 \sin \theta)^2 / (2g)$$

WORKED EXAMPLE — LAUNCH OFF A CLIFF

A ball is launched at $V_0=10$ m/s, 25° above horizontal, from the edge of a 1.5 m cliff. Find the max height, flight time, range, and impact velocity. Use $g=9.8$ m/s².

1. Components.

$$V_{0x} = 10 \cos 25^\circ = 9.06 \text{ m/s} \quad V_{0y} = 10 \sin 25^\circ = 4.23 \text{ m/s}$$

2. Max height above the launch point — apex is still reached before the ball ever gets near the base of the cliff, so the level-ground formula applies unchanged:

$$d_{y,\max} = (10\sin 25^\circ)^2 / (2 \cdot 9.8) = 17.86/19.6$$

$$d_{y,\max} \approx 0.91 \text{ m above the launch point}$$

3. **Flight time** — here the cliff matters. Launch and landing are *not* at the same height, so $t_{\text{flight}} = 2V_{0y}/g$ doesn't apply directly. Instead, write the full position equation, measuring from the launch point (landing is 1.5 m *below* it, so $d_y = -1.5$ m there):

$$d_y = d_{0y} + V_{0y}t - \frac{1}{2}gt^2 \Rightarrow -1.5 = (10\sin 25^\circ)t - 4.9t^2$$

$$4.9t^2 - 4.23t - 1.5 = 0 \quad (\text{quadratic formula, keep the positive root})$$

$$t_{\text{flight}} \approx 1.13 \text{ s}$$

4. **Range** — horizontal motion is constant-velocity the whole flight:

$$d_{x,\max} = V_{0x} \cdot t_{\text{flight}} = (9.06)(1.13)$$

$$d_{x,\max} \approx 10.2 \text{ m}$$

5. **Impact velocity** — recompute both components at $t = t_{\text{flight}}$:

$$V_x = V_{0x} = 9.06 \text{ m/s (unchanged)} \quad V_y = V_{0y} - gt = 4.23 - 9.8(1.13) \approx -6.85 \text{ m/s}$$

$$\text{speed} = \sqrt{(V_x^2 + V_y^2)} = \sqrt{(9.06^2 + 6.85^2)}$$

$$\text{impact speed} \approx 11.4 \text{ m/s, about } 52.9^\circ \text{ from vertical } (\approx 37^\circ \text{ below horizontal})$$

The impact velocity's angle can be measured from either axis — Dr. Cho's original work reports it from the vertical (52.9°), which is the same physical direction as about 37° below horizontal. Either description is correct as long as you say which axis you're measuring from; the Instructor Key has a note on this.

Turning the problem around: solving for θ

The worked example gave us V_0 , θ , and the cliff height H , and asked how far the ball goes. What if instead we're given V_0 , H , and a target distance D — and need to find the angle θ that lands exactly on it? Start from the same vertical position equation, but now solve for time using the horizontal equation first:

$$d_x = (V_0 \cos \theta)t = D \Rightarrow t = D / (V_0 \cos \theta)$$

$$0 = H + (V_0 \sin \theta)t - \frac{1}{2}gt^2 \Rightarrow 0 = H + D \cdot \tan \theta - gD^2 / (2V_0^2 \cos^2 \theta)$$

Using $1/\cos^2 \theta = 1 + \tan^2 \theta$ turns every θ in the equation into $\tan \theta$, which means this is really a **quadratic equation in $\tan \theta$** . Let $x = \tan \theta$:

$$0 = [H - gD^2 / (2V_0^2)] + D \cdot x - [gD^2 / (2V_0^2)] \cdot x^2$$

Solve this with the quadratic formula for x , then $\theta = \tan^{-1}(x)$. A quadratic has (up to) two roots, so this equation typically hands back **two launch angles** that both reach the same target — one a high, arcing lob and one a lower, flatter shot.

Check it yourself: for $H=1.2$ m, $D=1.5$ m, $V_0=5$ m/s, the bracketed quantities work out to $gD^2/(2V_0^2)=0.441$ and $H-0.441=0.759$, giving $0=0.759+1.5x-0.441x^2$. That solves to $x=3.848$ or $x=-0.447$, i.e. $\theta \approx 75.4^\circ$ or $\theta \approx -24.1^\circ$. (A negative angle here means the target is close enough that even aiming slightly downward from the cliff still reaches it.)

Looking ahead: N5 picks this derivation up in reverse — given where you want to land, it walks through solving that same $\tan \theta$ quadratic step by step for a specific launch problem. Everything you need for that is already sitting in the boxed equation above.

Exit ticket

Four quick questions — check yourself before you close this note.

Q1. Which velocity component determines how long a projectile stays in the air?

☐ Only the horizontal component, V_x .

☐ Only the vertical component, V_y .

☐ Both components equally.

☐ Neither — flight time is fixed by g alone.

Q2. At the apex of the trajectory...

☐ Both V_x and V_y are zero.

☐ $V_y = 0$, and V_x still equals $V_0 \cos\theta$.

☐ $V_x = 0$, and V_y is at its maximum.

☐ Both components are at their maximum.

Q3. A ball launches at $V_0 = 10$ m/s, 25° above horizontal, from a 1.5 m cliff ($g = 9.8$ m/s²). What is the flight time?

s

Q4. The general cliff-launch landing equation (given V_0 , H , and target distance D) turns out to be a quadratic in...

☐ θ itself.

☐ $\tan\theta$.

☐ V_0 .

☐ D .

Check my answers

Clear

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

Print or open on a tablet and work through the derivations and worked example as you follow along.

AFTER THE MODULE

Completed notes (PDF)

Released when Module 2 closes — check your work against it.